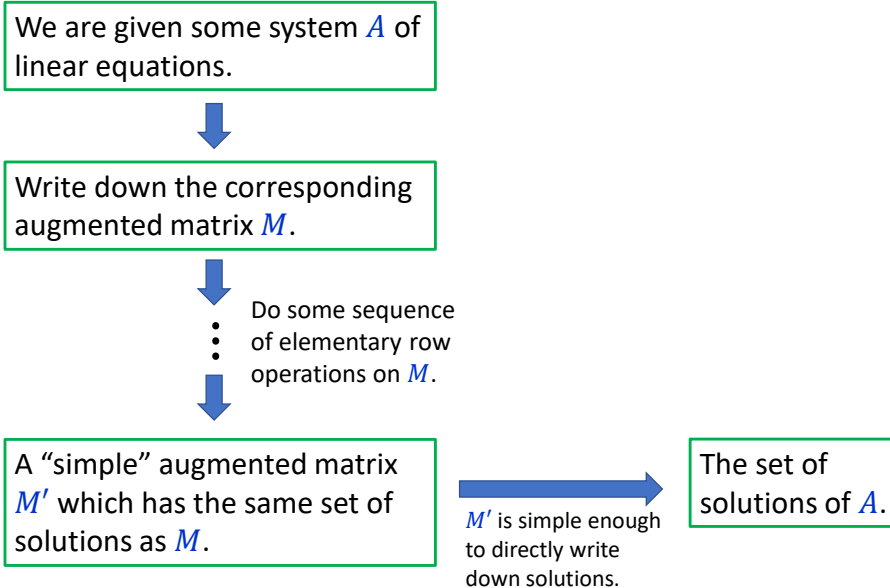
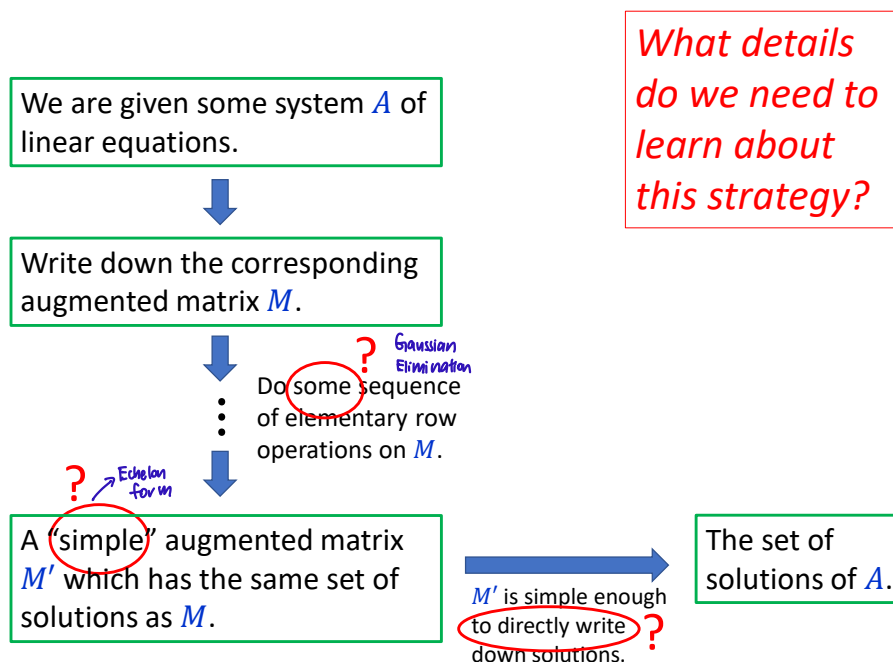


Echelon forms

Video 2.2.1

Recall our strategy for solving systems of linear equations:





We have two main questions to answer about this:

1. What do we mean by a “simple” matrix?

We will meet two important definitions.

- *A matrix in **row echelon form**.*
- *A matrix in **reduced row echelon form**.*

2. What is the sequence of elementary row operations which will turn a matrix into one of these “simple” forms?

We will meet two important algorithms which give a sequence of elementary row operations which will turn a matrix into one of these two “simple” forms.

Video 2.2.2

Row Echelon form

The first “simple” form of a matrix we will meet is called **Row Echelon form** of a matrix.

Definition (Row Echelon form of a matrix.)

Let M be a matrix or an augmented matrix.

We say that M is in **Row Echelon form** if it satisfies the following two rules:

1. If M has any rows that are entirely zeros then they all appear together right at the bottom of the matrix.
2. Given any two successive rows of M neither of which is all zeros, the first non-zero entry in the lower row (this is called the **leading entry** of the row) appears **strictly further to the right** than the first non-zero entry in the higher row.

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

↓ R_2 swap with R_3

$$\begin{bmatrix} 1 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \rightarrow \text{Row echelon form}$$

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 6 & -10 & 100 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

→ leading entry of row 1
→ leading entry of row 2
→ leading entry of row 3
→ this matrix is in row echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 10 \end{bmatrix}$$

→ this is the cause
→ not in row echelon form

Rule 1 says:

“If M has any rows that are entirely zeros then they all appear together right at the bottom of the matrix.”

In other words the matrix must either be entirely zero or must look like this:

$$\begin{bmatrix} * & * & * \\ \vdots & \vdots & \vdots \\ * & * & * \\ 0 & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & 0 \end{bmatrix}$$

None of these rows at the top are entirely zero.

All of the rows at the bottom are entirely zero.

For example:

$$\begin{bmatrix} 1 & 2 & 3 & -2 \\ 0 & 5 & 6 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Rule 2 says:

“Given any two successive rows neither of which is all zeros, the first non-zero entry in the lower row (this is called the **leading entry** of the row) appears **strictly further to the right** than the first non-zero entry in the higher row.”

In other words, for any two successive non-zero rows, the picture must be the following:

$$\begin{array}{l} \Rightarrow \\ \Rightarrow \end{array} \begin{bmatrix} 0 & \cdots & 0 & \neq 0 & * & * & * & * & * \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \neq 0 & * & * \end{bmatrix}$$

The circled entries are the **leading entries** of these two rows.

This row is further down the matrix than the other highlighted row, so its leading entry must be strictly further to the right.

So the typical shape of “row echelon form” is the following upside-down staircase...

$$\left(\begin{array}{cccc|ccc} \otimes & * & & & & * & & \\ & \otimes & * & & & * & & \\ & & \ddots & \ddots & & * & & \\ & & & \otimes & * & * & & \\ 0 & \dots & & & & 0 & 0 & \end{array} \right) \begin{array}{l} \left. \vphantom{\begin{array}{cccc|ccc} \otimes & * & & & & * & & \\ & \otimes & * & & & * & & \\ & & \ddots & \ddots & & * & & \\ & & & \otimes & * & * & & \end{array}} \right\} \text{nonzero rows} \\ \left. \vphantom{\begin{array}{cccc|ccc} 0 & \dots & & & & 0 & 0 & \end{array}} \right\} \text{zero rows (if any)} \end{array}$$

leading entries

I wonder where this word Echelon comes from... Let's google it...

Quick check of understanding:

Video 2.2.3

Are any of the following matrices in row echelon form?

(a)
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
 No. The row of zeros is supposed to be in the last row

(b)
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
 Yes.

(c)
$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 0 & 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 10 & 100 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$
 No. The position of the leading entry did not change from row 2 to row 3.

Solving a system whose augmented matrix is in Row Echelon form

If you have a system of linear equations whose augmented matrix is in row echelon form, then you can calculate its set of solutions directly using a straightforward procedure called **back-substitution**.

Back-substitution.

Video 2.2.4

You'll find back-substitution is very simple to do in practice. We'll introduce it in a minute by an example, but here are the key ideas that we'll meet:

- Solve the equations one-by-one, starting with the last equation first and then proceeding from there up the list.
- At each step, you are using the equation corresponding to some row to solve for the variable corresponding to the leading entry of that row.
- At each step, substitute the variables you have solved earlier in the process into the equation, and set any remaining variables besides the leading entry to be free parameters.

Introductory Example of Back-substitution #1

Let's start with a super simple example.

Consider the following system of linear equations:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ \quad 4x_2 + 5x_3 = 1 \\ \quad \quad 2x_3 = 6 \end{cases} \quad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 4 & 5 & 1 \\ 0 & 0 & 2 & 6 \end{array} \right]$$

Exercise: Compute the set of solutions.

Example (cont.)

Example (cont.)

Consider the following system of linear equations:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ \quad 4x_2 + 5x_3 = 1 \\ \qquad \quad 2x_3 = 6 \end{cases}$$

- We start with the last equation. This equation is true precisely when $x_3 = 3$.
- Then we move up the list to the second equation. We already know what x_3 must be so we can immediately substitute that into the second equation. The second equation becomes $4x_2 + 5 \cdot 3 = 1$. This equation is true precisely when $x_2 = -7/2$.
- Finally we reach the first equation. We already know what x_2 and x_3 must be so we can substitute the answers in immediately. The first equation becomes $x_1 + 2 \cdot \left(-\frac{7}{2}\right) + 3 \cdot 3 = 0$. This equation is true precisely when $x_1 = -2$.

Example. (Discussion.)

Consider the following system of linear equations:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 0 \\ \quad 4x_2 + 5x_3 = 1 \\ \qquad \quad 2x_3 = 6 \end{cases}$$

In summary, this system has **exactly one solution**: $\begin{cases} x_1 = -2 \\ x_2 = -7/2 \\ x_3 = 3 \end{cases}$.

This solution was very simple precisely because the augmented matrix corresponding to this system was in **row echelon form**:

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 4 & 5 & 1 \\ 0 & 0 & 2 & 6 \end{array} \right].$$

As we move from the bottom equation up the matrix, at every step there is a new variable (the leading term) which we can express in terms of what we have solved already.

The general situation for back-substitution

Video 2.2.5

The general situation of a system in Echelon form is more complicated than the previous example.

The reason is: as we go up the matrix it is possible that some row introduces **several** variables which we **haven't solved** for already. An example might be the system:

$$\begin{cases} x_1 + 2x_2 + x_3 + 2x_4 = 1 \\ \quad \underline{x_2} + \underline{x_3} + 3x_4 = 2 \\ \qquad \qquad \qquad 5x_4 = 10 \end{cases} \quad \leftarrow$$

But this is still easy: solve for the left-most variable like before, but now make any remaining new variables **free parameters**.

For example in this system: The bottom equation tells you $x_4 = 2$. Then the next equation becomes $x_2 + x_3 + 3 * 2 = 2$ which is the same as $x_2 + x_3 = -4$. This has 2 variables we haven't see before. You set x_3 to a new free variable $x_3 = s$ and then you solve for x_2 in terms of $x_3 = s$ to get $x_2 = -4 - s$.

Let's do this more systematically using an example.

Introductory Example of Back-substitution #2

Consider the following augmented matrix. Observe that it is in row echelon form:

$$\left[\begin{array}{ccccc|c} 0 & 2 & 2 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 4 \end{array} \right]$$

Each column corresponds with a variable of the associated linear system. Let's assume they are $x_1, \dots, x_5$:

$$\begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \\ \left[\begin{array}{ccccc|c} 0 & 2 & 2 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 4 \end{array} \right] \end{array}$$

Example (cont.)

Each column corresponds with a variable of the associated linear system. Let's assume they are $x_1, \dots, x_5$:

$$\begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \\ \left[\begin{array}{ccccc|c} 0 & 2 & 2 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 4 \end{array} \right] \end{array}$$

The corresponding linear system is:

$$\begin{aligned} 2x_2 + 2x_3 + x_4 - 2x_5 &= 2 \\ x_3 + x_4 + x_5 &= 3 \\ 2x_5 &= 4 \end{aligned}$$

The first equation is solved last.

Then you go up the list.

In Back-substitution you solve the last equation first.

Example (cont.)

At each step you are solving for the variable corresponding to the leading entry....

$$\begin{array}{ccccc} x_1 & x_2 & x_3 & x_4 & x_5 \\ \left[\begin{array}{ccccc|c} 0 & 2 & 2 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 4 \end{array} \right] \end{array}$$

Free parameter $\nearrow$

You start with the last equation and solve it for variable x_5 .

Then you solve the second last equation for the variable x_3 using the answer you already found for x_5 and setting the skipped variable x_4 to a free parameter... and so on.

Example (cont.)

OK let's do it. The system we are solving is:

$$\begin{cases} 2x_2 + 2x_3 + x_4 - 2x_5 = 2 & (1) \\ x_3 + x_4 + x_5 = 3 & (2) \\ 2x_5 = 4 & (3) \end{cases}$$

We'll solve this from the bottom up, (3) then (2), then (1).

Equation (3) gives: $x_5 = 2$.

Then we substitute this into the next equation, equation (2).

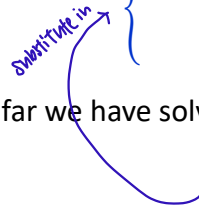
Equation (2) becomes (2'): $x_3 + x_4 + 2 = 3$. Or: $x_3 + x_4 = 1$.

Next, to solve (2') we let x_4 be a new free parameter $x_4 = s$ and solve for the leading term x_3 in terms of this new parameter: $x_3 = 1 - s$.

Example (cont.)

The system we are solving is:

$$\begin{cases} 2x_2 + 2x_3 + x_4 - 2x_5 = 2 & (1) \\ x_3 + x_4 + x_5 = 3 & (2) \\ 2x_5 = 4 & (3) \end{cases}$$

Substitute in 

So far we have solved the bottom two equations and found:

$$\begin{cases} x_5 = 2 \\ x_4 = s \\ x_3 = 1 - s \end{cases}, \quad s \in \mathbb{R}.$$

Finally we solve the first equation. We substitute these answers we found already into equation (1). We get:

$$2x_2 + 2(1 - s) + s - 2 * 2 = 2$$

We can solve this for x_2 : $x_2 = 2 + \frac{1}{2}s$.

Example (cont.)

The system we are solving is:

$$\begin{cases} 2x_2 + 2x_3 + x_4 - 2x_5 = 2 & (1) \\ x_3 + x_4 + x_5 = 3 & (2) \\ 2x_5 = 4 & (3) \end{cases}$$

So far we have solved all three equations and found:

$$\begin{cases} x_5 = 2 \\ x_4 = s \\ x_3 = 1 - s, \\ x_2 = 2 + \frac{1}{2}s \end{cases}, \quad s \in \mathbb{R}.$$

Are we done? Well we started with 5 columns. What about x_1 ?

There are no equations using x_1 , so x_1 can be anything. We set it to a new free parameter $x_1 = t$.

Example (cont.)

The system we are solving is:

$$x_1 = t \quad \begin{cases} 2x_2 + 2x_3 + x_4 - 2x_5 = 2 & (1) \\ x_3 + x_4 + x_5 = 3 & (2) \\ 2x_5 = 4 & (3) \end{cases}$$

The general solution of this system is:

$$\begin{cases} x_5 = 2 \\ x_4 = s \\ x_3 = 1 - s \\ x_2 = 2 + \frac{1}{2}s \\ x_1 = t \end{cases}, \quad s, t \in \mathbb{R}.$$

There are **two free parameters** so there are infinitely many solutions. (To be precise we will see later that the set of solutions is a “2-dimensional affine subspace of $\mathbb{R}^5$ ”.)

Example (cont.)

Let's summarise some key features of this calculation.
Here was the augmented matrix we started with:

We used the equation corresponding to a row to solve for the variable corresponding to the leading entry of the row. E.g. row 2 was used to solve for x_3 .

$$\begin{array}{ccccc|c}
 x_1 & x_2 & x_3 & x_4 & x_5 & \\
 \hline
 0 & 2 & 2 & 1 & -2 & 2 \\
 0 & 0 & 1 & 1 & 1 & 3 \\
 0 & 0 & 0 & 0 & 2 & 4
 \end{array}$$

Variables that did not correspond to leading entries became free parameters.

Quick test of understanding

Video 2.2.6

If you have a linear system whose corresponding augmented matrix is the following, how many free parameters do you expect in a parameterization for the set of solutions?

$$\begin{array}{cccccccc|c}
 0 & 0 & 1 & 8 & 2 & 2 & 1 & -2 & 0 & 2 \\
 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 3 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 4 \\
 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
 \end{array}$$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow \quad \uparrow$
 free parameters

Answer:

There are 3 non-zero rows, so there are 3 leading entries in 3 different columns. They are marked in pink. The 3 equations are used to solve for these variables. The remaining 6 columns will correspond to free parameters. Thus there will be 6 free parameters in a parameterization of the set of solutions.

Exercise

Video 2.2.7

You are given a linear system with variables $x_1, \dots, x_5$.

The augmented matrix of the linear system is:

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 2 \\ 0 & 1 & 0 & 1 & 1 & 30 \\ 0 & 0 & 0 & 1 & 1 & 10 \end{array} \right].$$

Compute a parameterization of the set of solutions of this system.

Exercise

You are given a linear system with variables $x_1, \dots, x_5$.

The augmented matrix of the linear system is:

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & -2 & 2 \\ 0 & 1 & 0 & 1 & 1 & 30 \\ 0 & 0 & 0 & 1 & 1 & 10 \end{array} \right].$$

Compute a parameterization of the set of solutions of this system.

Solution

The corresponding system is:

$$\begin{cases} x_1 & -2x_5 & = & 2 \\ x_2 & +x_4 & +x_5 & = & 30 \\ & x_4 & +x_5 & = & 10 \end{cases}$$

Solution (cont.)

We start by solving the third equation. The left-most variable in this equation is x_4 and we will solve for that variable. The extra variable we haven't solved for already is x_5 which we will set to a new free parameter $x_5 = r$. Thus the bottom equation gives us:

$$\begin{cases} x_4 = 10 - r \\ x_5 = r \end{cases}, r \in \mathbb{R}$$

Solution (cont.)

Next, we move up the system to the second equation. We will solve for the left-most variable (in this case x_2) while substituting in the solution we have already.

We get: $x_2 = 30 - (10 - r) - r = 20$

So the solution so far becomes:

$$\begin{cases} x_2 = 20 \\ x_4 = 10 - r \\ x_5 = r \end{cases}, r \in \mathbb{R}$$

Solution (cont.)

Next, we move up the system to the first equation. We will solve for the left-most variable (in this case x_1) while substituting in the solution we already have. We get: $x_1 = 2 + 2r$

Finally note that there is a variable we haven't solved for yet: x_3 . Actually, this variable doesn't appear in any equations, so we can set it to anything at all. It becomes a new free parameter: $x_3 = s$

Solution (Conclusion.)

The set of solutions has the following parameterization:

$$\begin{cases} x_1 = 2 + 2r \\ x_2 = 20 \\ x_3 = s \\ x_4 = 10 - r \\ x_5 = r \end{cases}, \quad r, s \in \mathbb{R}.$$

There is one more possibility we need to mention:

Video 2.2.8

Introductory Example of Back-substitution #3

Consider the following augmented matrix.

$$\left[\begin{array}{cc|c} 2 & 2 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 4 \end{array} \right]$$

This matrix is in row echelon form.

We can see immediately that it has **no solutions**. In other words it is **inconsistent**.

The reason is that the equation corresponding to the third row is: $0x_1 + 0x_2 = 4$. There are **no choices for x_1 and x_2** that will make this equation true because the LHS is always 0.

Where we stand

Start with a linear system.

Write down the augmented matrix.

This step is easy



Repeatedly do elementary row operations on the matrix until it is "simple".
(We'll see shortly there are TWO patterns we'll consider "simple").



When the augmented matrix is in one of these "simple" forms, we can solve for the variables. We were just learning how to do this step by BACK-SUBSTITUTION

Gaussian Elimination

Video
2.2.9

An algorithm for transforming an augmented matrix of a linear system into row-echelon form by elementary row operations

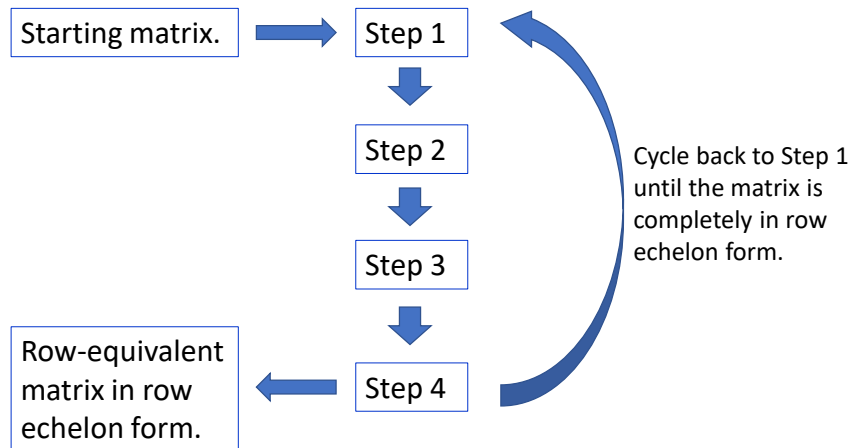
- Elementary row operations don't change the solution set of the linear system
- When the augmented matrix is in row-echelon form the system can be easily solved by back-substitution.
- Therefore, Gaussian elimination allows us to solve the linear system.

By the way:

Gaussian elimination is one of the two or three most important things in this course.

The Algorithm called **Gaussian Elimination**.

This algorithm performs elementary row operations in a cycle of 4 steps which we will repeat until the algorithm terminates when the matrix is in row echelon form.



The Algorithm called **Gaussian Elimination**.

We'll introduce this algorithm by following through the two examples below.

Step 1:

Locate the **left-most column** that is **not entirely zeros**.

$$\begin{bmatrix} 0 & 3 & \cdots & \cdots \\ 1 & -2 & \cdots & \cdots \\ 4 & 0 & \cdots & \cdots \end{bmatrix}$$

↑
the first
nonzero
column

$$\begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 2 & -3 & \cdots \\ 0 & -1 & 6 & \cdots \end{bmatrix}$$

↑
the first
nonzero
column

The Algorithm called **Gaussian Elimination**. (cont.)

Step 2:

If necessary swap the top row (call it R_T) with another row (using the elementary row operation $R_T \leftrightarrow R_B$) so as to ensure that the top entry of the column you identified in Step 1 is not a zero.

In this case, this is a zero so we need to replace the top row with another row.

For example, you could swap the 1st row with the 2nd row using $R_1 \leftrightarrow R_2$.

$$\begin{bmatrix} 0 & 3 & \cdots & \cdots \\ 1 & -2 & \cdots & \cdots \\ 4 & 0 & \cdots & \cdots \end{bmatrix}$$

In this case the top entry of the first non-zero column is not zero, so no action is needed.

$$\begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 2 & -3 & \cdots \\ 0 & -1 & 6 & \cdots \end{bmatrix}$$

The Algorithm called **Gaussian Elimination**. (cont.)

Step 3:

Let the top row be called R_T . For every row below the top row, call it R_S , add a suitable multiple of the top row R_T to it (via some choice of elementary row operation $R_S \rightarrow R_S + cR_T$ for some $c \in \mathbb{R}$) so that the entry of R_S immediately below the leading entry of R_T becomes zero.

$$\begin{bmatrix} 1 & -2 & \cdots & \cdots \\ 0 & 3 & \cdots & \cdots \\ 4 & 0 & \cdots & \cdots \end{bmatrix}$$

$$\begin{bmatrix} 0 & 3 & 1 & \cdots \\ 0 & 2 & -3 & \cdots \\ 0 & -1 & 6 & \cdots \end{bmatrix}$$

Add -4 times of the 1st row to the 3rd row so that the entry marked by ● becomes 0.

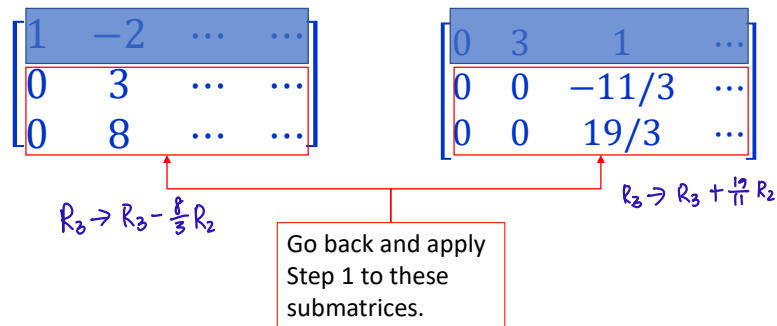
Add $-2/3$ times of the 1st row to the 2nd row so that the entry marked by ● becomes 0.

Add $1/3$ times of the 1st row to the 3rd row so that the entry marked by ● becomes 0.

The Algorithm called **Gaussian Elimination**. (cont.)

Step 4:

Now cover the top row in the matrix and begin again with Step 1 applied to the submatrix that remains.



Repeat these steps until you reach the last row.

Example:

Video 2.2.10

Consider the following augmented matrix. We will apply Gaussian elimination to obtain a row equivalent matrix in row echelon form.

$$\left[\begin{array}{ccccc|c} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right]$$

Step 1: The first column is the left-most non-zero column.

Step 2: The top entry of the column we identified in Step 1 (namely, column 1) is a zero. So we have to do a row swap to fix that. We can do the swap $R_1 \leftrightarrow R_2$.

Example (continued):

After the row swap we get:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right]$$

Step 3: We want to make all the entries under the top entry of the first non-zero column zero. There is only one entry we need to fix, the entry circled. To make that zero we can replace R_3 by $R_3 + 2R_1$. We get:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{array} \right]$$

Example (continued):

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{array} \right]$$

Step 4: Cover the top row of the matrix, and go back to Step 1.

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{array} \right]$$



Step 1 (Second time around): The first non-zero column is now column 3.

Step 2 (Second time around): The top entry of the first non-zero column is not zero, so we don't need any row swaps.

Example (continued):

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{array} \right]$$


Step 3 (Second time around): We want to make all the entries under the top entry of the first non-zero column into zeros. So we want to replace this 3 by a zero. The elementary row operation $R_3 \rightarrow R_3 - \frac{3}{2}R_2$ does that.

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right]$$

Example (continued):

Step 4 (Second time around): We cover the top row.

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right]$$

We have reached the bottom row of the matrix, so the algorithm is concluded. The following is an augmented matrix in row echelon form which has been obtained from our original matrix by elementary row operations:

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right]$$

Example (summary):

$$\left[\begin{array}{ccccc|c} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right]$$

The augmented matrix we started with.



$$\begin{aligned} R_1 &\leftrightarrow R_2 \\ R_3 &\rightarrow R_3 + 2R_1 \\ R_3 &\rightarrow R_3 - \frac{3}{2}R_2 \end{aligned}$$

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right]$$

This is a row equivalent augmented matrix in row echelon form.

Example 1.4.9.4

Given a **cubic curve** with equation

$$y = a + bx + cx^2 + dx^3,$$

where a, b, c, d are real constants, that passes through the points

$$(0, 10), (1, 7), (3, -11) \text{ and } (4, -14),$$

find the values of a, b, c, d .

Example 1.4.9.4 Solution

By substituting $(x, y) = (0, 10)$, $(1, 7)$, $(3, -11)$ and $(4, -14)$ into the equation $y = a + bx + cx^2 + dx^3$, we obtain a system of linear equations:

$$\begin{cases} a & = & 10 \\ a + b + c + d & = & 7 \\ a + 3b + 9c + 27d & = & -11 \\ a + 4b + 16c + 64d & = & -14 \end{cases}$$

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This method is usually named for:

Carl Friedrich Gauss (1777-1855)



Carl Friedrich Gauss, a German scientist known as the **Prince of Mathematicians**, is generally regarded as one of the greatest mathematicians of all time for his contributions to number theory, geometry, probability theory, geodesy, planetary astronomy, the theory of functions, and potential theory.

But maybe we should also Google some history...